

Swahili Proverb Classification

Application of Natural Language Processing Classification Methods for Low-Resource and Agglutinative Languages

Submitted by

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For

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1. Introduction

"Haba na haba hujaza kibaba" is a Swahili proverb meaning that small and consistent efforts accumulate into meaningful outcomes. This proverb reflects a working principle in machine learning: models improve progressively and incrementally through exposure to data and better representations. Natural language processing (NLP), as a branch of machine learning concerned with how computers represent and interpret human language, embodies this same principle. Progress on a task like Swahili proverb classification depends significantly on how well the underlying text is represented, and these representations improve incrementally as methods advance from simple word counts to richer contextual models.

Swahili is one of Africa's most widely spoken languages, serving as a lingua franca across East and Central Africa with an estimated 200 million speakers (Ethnologue, 2023). Despite this wide reach, Swahili remains significantly under-represented in NLP research, with fewer dedicated resources and benchmark datasets compared to high-resource languages such as English, French, or Mandarin. Proverbs constitute a particularly rich form of Swahili language use: they encode accumulated wisdom through concise, metaphorical expressions that bear no surface-level semantic resemblance to their underlying meaning. A proverb categorised under 'Debt/Borrowing' does not typically contain the Swahili word for borrowing; instead, it may describe a falling tree, a setting sun, or a broken bag. This semantic gap makes proverb classification a test of a model's capacity for deep semantic understanding.

The challenge is compounded by Swahili's agglutinative morphology. Unlike English, where grammatical information is largely conveyed through separate function words, Swahili encodes tense, subject, object, and mood through affixes attached to a single verb stem, so a single Swahili token can carry the semantic load of an entire English phrase. This paper is a comparative study of four NLP methods: classical static embedding methods (TF-IDF, Word2Vec, FastText), transformer fine-tuning (DistilBERT-multilingual, SwahBERT, AfriBERTa-Large), MLM domain adaptation, and large language model prompting (Llama-3-8B-Instruct). The study addresses four research questions:

1. Which performs best on this low-resource, metaphorically rich classification task?
2. Does language-specific pretraining achieve a measurable advantage over multilingual models?
3. Does MLM domain adaptation on the target corpus provide further improvement?
4. Can a large language model perform competitively without task-specific training, under zero-shot and few-shot conditions?

2. Related Works

Text classification has evolved from sparse statistical representations (Sparck Jones, 1972) through dense distributed embeddings (Mikolov et al., 2013; Bojanowski et al., 2017) to deep contextual representations introduced by BERT (Devlin et al., 2019) and its derivatives including DistilBERT (Sanh et al., 2019). For specialised domains, domain-adaptive pretraining (Gururangan et al., 2020), continued MLM pretraining on the target domain before fine-tuning, has been established as a reliable low-resource technique. Large language models such as Llama-3 (Dubey et al., 2024) have introduced in-context learning as a new classification paradigm: providing task descriptions and optional examples in the prompt, without gradient updates. Few-shot prompting has achieved

competitive results on many benchmarks (Brown et al., 2020), though performance on low-resource languages and culturally specific tasks remains less well characterised.

Research on NLP for African languages has expanded considerably in recent years. AfriBERTa (Ogueji et al., 2021) demonstrated that competitive multilingual language models can be trained on less than 1 GB of text across 11 African languages including Swahili, challenging the assumption that large pretraining corpora are a prerequisite for effective language models. AfroXLMR (Alabi et al., 2022) showed that multilingual adaptive fine-tuning of multilingual models on African language corpora substantially improves downstream task performance, suggesting that language and domain adaptation can partially compensate for a pretraining distribution dominated by high-resource languages. SwahBERT (Martin et al., 2022) provides a monolingual BERT for Swahili trained on news, online discussion, and Wikipedia text, motivating a direct comparison with multilingual alternatives: a dedicated monolingual model should better capture Swahili's morphological and syntactic patterns, though this advantage may diminish on a domain as semantically unusual as proverbs. To our knowledge, no prior study has systematically compared this full spectrum of NLP approaches on Swahili proverb classification.

3. Methodology

Dataset Description

The dataset consists of 5,060 Swahili proverbs organised into 56 semantic categories, sourced from the Swahili Proverbs collection of the Center for African Studies at the University of Illinois Urbana-Champaign (Scheven, 2010). Categories include abstract thematic labels such as Friendship, Debt/Borrowing, Work and Laziness, Poverty and Wealth, Speech and Silence, and Pride and Humility. The proverbs average approximately 5 to 9 words and 30 characters in length. The dataset shows significant class imbalance: the Alertness category is the most represented while Drinking/Eating contains the fewest entries, with some categories having as few as 5 examples while others exceed 200. Data was split using stratified sampling with a fixed random seed (seed=42), ensuring reproducibility and balanced class representation, as shown in **Table 1**.

Split	Samples	Percentage
Training	3,643	~70%
Validation	405	~10%
Test	1,012	~20%
Total	5,060	100%

Table 1. Dataset split statistics.

Data Preprocessing

Minimal text processing was carried out given the average length of the proverbs. No stop words were removed; all text was lowercased and tokenised using whitespace and punctuation boundaries. No stemming or lemmatisation was applied: given Swahili's agglutinative morphology, aggressive stemming risks stripping meaningful grammatical affixes, such as tense prefixes *ali-* (past) and *ata-* (future), and noun class prefixes *m-*, *wa-*, *ki-*, *vi-*, that may carry discriminative signal for certain semantic categories. For transformer models, tokenisation was handled by each model's own

pretrained tokenizer with a maximum sequence length of 64 tokens, which comfortably accommodates all proverbs in the corpus.

Evaluation Metrics

Given the significant class imbalance across the 56 categories, standard accuracy is a potentially misleading metric: a model predicting only majority classes can achieve high accuracy while failing on most underrepresented categories. Macro-F1 and Matthews Correlation Coefficient (MCC) are therefore adopted as supplementary metrics (Chicco and Jurman, 2020). Test accuracy is also reported as the primary metric across all tasks to enable consistent cross-task comparison.

Reproducibility and Tools

All experiments were implemented in Python and run in a Google Colab GPU environment. Classical baselines used scikit-learn for the TF-IDF and character n-gram vectorisers, train/validation/test splitting, and metric computation; Word2Vec embeddings were trained with gensim. Neural training used PyTorch, and all transformer and language models were loaded through the Hugging Face Transformers library. A fixed random seed of 42 was applied to Python, NumPy, and PyTorch throughout to support reproducibility. All training and evaluation runs were tracked with Weights & Biases and MLflow.

Task 1: Static Embeddings + MLP

Task 1 establishes baselines using static text representations combined with a shallow neural classifier. Five representation methods are evaluated, summarised in **Table 2**. TF-IDF (1-2gram) is a strong baseline for short-text classification because it down-weights common tokens and up-weights discriminative ones, although it is blind to morphological structure and cannot generalise to out-of-vocabulary words. Character n-gram TF-IDF applies TF-IDF at the character level (n-grams of length 3 to 6), which is well-motivated for Swahili because character n-grams implicitly capture morphological affixes. Word2Vec trains skip-gram embeddings of dimension 100 on the training corpus, with 1-epoch and 10-epoch variants to assess whether additional iterations yield more stable representations; each proverb is represented as the mean of its token embeddings. FastText uses Facebook's pre-trained Swahili Common Crawl and Wikipedia vectors (cc.sw.300.bin), providing 300-dimensional embeddings with subword coverage (Grave et al., 2018). The classifier is a two-layer SimpleMLP with hidden dimension 128, ReLU activation, and cross-entropy loss, trained with Adam (lr=0.001, batch size=32) for 50 epochs. The checkpoint with highest validation Macro-F1 is retained for test evaluation.

Method	Type	Dimension	Pretrained on Swahili?
TF-IDF (1-2gram)	Sparse bag-of-words	1,000	No
Char N-gram TF-IDF	Sparse char n-grams (3 to 6)	2,000	No
Word2Vec (1 ep)	Dense word embeddings	100	No
Word2Vec (10 ep)	Dense word embeddings	100	No
FastText (cc.sw.300)	Dense subword embeddings	300	Yes

Table 2. Task 1 representation methods.

Task 2: Transformer Fine-Tuning

Task 2 fine-tunes transformer models for sequence classification, each with a classification head mapping the [CLS] token representation to 56 output logits, trained with AdamW, linear warmup, and gradient clipping at norm 1.0 following standard BERT fine-tuning practice (Devlin et al., 2019). The configurations are described in **Table 3**. DistilBERT-multilingual (distilbert-base-multilingual-cased) was run at both 3 and 10 epochs at $\text{lr}=3\text{e-}5$ to observe epoch sensitivity. SwahBERT (Martin et al., 2022), a monolingual Swahili BERT, was fine-tuned for 5 epochs at $\text{lr}=2\text{e-}5$; its vocabulary and tokeniser are natively designed for Swahili morphology, although proverbs are stylistically distant from the news-domain text that dominated its pretraining. AfriBERTa-Large (castorini/afriberta_large), a RoBERTa-style model pretrained on 11 African languages including Swahili (Ogueji et al., 2021), was fine-tuned for 5 epochs at $\text{lr}=2\text{e-}5$.

Model	Architecture	Pretraining	Epochs	LR
DistilBERT-multilingual	6-layer distilled BERT	104 languages	3 / 10	$3\text{e-}5$
SwahBERT	12-layer BERT	Swahili (monolingual)	5	$2\text{e-}5$
AfriBERTa-Large	RoBERTa-style	11 African languages	5	$2\text{e-}5$

Table 3. Task 2 model configurations.

Task 3: MLM Domain Adaptation

Task 3 investigates whether continued MLM pretraining on the proverb corpus improves downstream classification, following the domain-adaptive pretraining paradigm (Gururangan et al., 2020). The full 5,060 proverb corpus is used for MLM adaptation (training, validation, and test combined), as labels are not used. This is a self-supervised procedure: 15% of tokens are dynamically masked per batch using Hugging Face's DataCollatorForLanguageModeling, and the model is trained to recover the original tokens. Adaptation runs for 1 epoch (AdamW, $\text{lr}=5\text{e-}5$, gradient clipping 1.0). AfriBERTa-Large and SwahBERT are each evaluated at baseline and post-adaptation, producing the four conditions shown in **Table 4**.

Condition	Base Model	MLM Adapted?	Cls Epochs
AfriBERTa-Baseline	castorini/afriberta_large	No	5
AfriBERTa-MLM	castorini/afriberta_large	Yes (1 ep, $\text{lr}=5\text{e-}5$)	5
SwahBERT-Baseline	SwahBERT	No	5
SwahBERT-MLM	SwahBERT	Yes (1 ep, $\text{lr}=5\text{e-}5$)	5

Table 4. Task 3 experimental conditions.

Task 4: LLM Prompting with Llama-3

Task 4 evaluates Llama-3-8B-Instruct (Dubey et al., 2024) under zero-shot and few-shot prompting on a random sample of 20 proverbs. For zero-shot classification, each proverb is presented with all 56 category names and a system prompt instructing the model to respond with only the category name, using the Llama-3 chat template. For few-shot classification, 10 randomly selected categories are each illustrated with one example proverb before the target is presented. Label extraction uses cascaded matching: exact string match, then substring match, then word-level overlap. Predictions matching no category are counted as invalid and excluded from metric computation. Because of the small sample, results at $n=20$ across 56 classes are necessarily noisy and should be treated as exploratory rather than definitive.

4. Results

This section reports results for each task. Test accuracy, available for every MODEL, is used as the common metric for cross-task comparison, with Macro-F1 reported where computed.

Task 1: Static Embeddings

Test accuracy results for all five static embedding methods are presented in **Table 5**. TF-IDF (1-2gram) achieved the highest accuracy at 0.240, followed by Char N-gram TF-IDF at 0.214. Both sparse representation methods substantially outperformed the dense embedding approaches. FastText (cc.sw.300), despite being pretrained on large-scale Swahili text, performed weakest at 0.047. Word2Vec trained for 10 epochs (0.055) underperformed the 1-epoch variant (0.076), suggesting that on this small corpus additional training epochs do not improve generalisation and may overfit the limited vocabulary.

Method	Test Accuracy
TF-IDF (1-2gram)	0.240
Char N-gram TF-IDF	0.214
Word2Vec (1 ep)	0.076
Word2Vec (10 ep)	0.055
FastText (cc.sw.300)	0.047

Table 5. Task 1 results, static embeddings + MLP classifier.

Task 2: Transformer Fine-Tuning

Results for the transformer fine-tuning experiments are shown in **Table 6**. AfriBERTa-Large achieved the highest test accuracy (0.202) and Macro-F1 (0.109) among all transformer configurations. SwahBERT ranked second with accuracy 0.155 and Macro-F1 0.066. DistilBERT-multilingual at 10 epochs reached accuracy 0.144 and Macro-F1 0.072, substantially outperforming the 3-epoch run (accuracy 0.069, Macro-F1 0.010), confirming that the multilingual model requires more fine-tuning to converge on this task. All four transformer configurations underperformed TF-IDF on test accuracy.

Model	Test Accuracy	Test Macro-F1
DistilBERT-multilingual (3ep)	0.069	0.010
DistilBERT-multilingual (10ep)	0.144	0.072
SwahBERT	0.155	0.066
AfriBERTa-Large	0.202	0.109

Table 6. Task 2 results, transformer fine-tuning.

Task 3: MLM Domain Adaptation

The effect of MLM domain adaptation is presented in **Table 7**. AfriBERTa-MLM produced a small positive gain over its baseline (+0.004 accuracy, from 0.202 to 0.206), making it the highest-performing transformer configuration overall. SwahBERT-MLM showed a slight decline relative to its baseline (−0.012 accuracy, from 0.155 to 0.143). The contrasting directions suggest that a single epoch of MLM adaptation may introduce noise at this corpus scale rather than consistently improving representations. Neither adapted model closed the gap with TF-IDF.

Condition	Test Accuracy	Test Macro-F1
AfriBERTa-Baseline	0.202	0.109
AfriBERTa-MLM	0.206	n/a
SwahBERT-Baseline	0.155	0.066
SwahBERT-MLM	0.143	n/a

Table 7. Task 3 results, MLM domain adaptation effect.

Task 4: Llama-3 Prompting

Results for Llama-3-8B-Instruct evaluated on 20 proverbs are shown in **Table 8**. Zero-shot prompting achieved test accuracy of 0.200 and few-shot prompting (10 illustrated categories) achieved 0.150. The few-shot result is lower than zero-shot, suggesting that the example demonstrations did not improve category alignment and may have biased predictions toward the 10 illustrated categories. Both conditions are treated as exploratory given the small evaluation set.

Method	Test Accuracy
Zero-shot	0.200
Few-shot (10 examples)	0.150

Table 8. Task 4 results, Llama-3-8B zero-shot and few-shot prompting.

Cross-Task Summary

Table 9 consolidate the best-performing method from each task. TF-IDF with SimpleMLP achieved the highest test accuracy at 0.240. AfriBERTa-Large with MLM adaptation was the best transformer result at 0.206. Llama-3 zero-shot reached 0.200. The narrow spread between the top transformer and LLM results (0.200 to 0.206) contrasts with the larger gap to TF-IDF (0.240), reinforcing the finding that statistical surface representations outperform contextual models on this small, metaphorically dense corpus.

Task	Best Method	Test Accuracy
Task 1: Static Embeddings	TF-IDF (1-2gram)	0.240
Task 2: Transformer Fine-tuning	AfriBERTa-Large	0.202
Task 3: MLM + Fine-tuning	AfriBERTa + MLM	0.206
Task 4: LLM Prompting	Llama-3 Zero-shot	0.200

Table 9. Best-performing method per task, cross-paradigm comparison.

5. Discussion

With 56 categories and support ranging from 5 to over 200 samples, standard accuracy can be deceptive. Macro-F1 directly penalises models that concentrate predictions on frequent categories, and MCC is more comprehensive (Chicco and Jurman, 2020). AfriBERTa-Large reaches 0.109 Macro-F1 against 0.202 accuracy, indicating that accuracy is inflated by correct predictions on high-support categories while Macro-F1 reveals poor coverage of the long tail. Per-class classification reports are provided in **Appendix B**.

The fundamental challenge of this task is the semantic gap between proverb text and category label. A proverb about a setting sun does not contain the word 'debt'; a proverb about a departing visitor does not contain the word 'friendship'. Models must learn to associate surface lexical and morphological patterns with abstract thematic categories purely from labelled examples, a form of metaphor comprehension that remains an open problem in NLP. The low absolute performance of all methods (best accuracy 0.240 on a 56-class problem, against a random baseline of roughly 0.018) reflects this difficulty.

TF-IDF's superiority over transformer models is consistent with this framing. With as few as 3 to 4 training examples per class in some categories, fine-tuning transformers does not provide sufficient signal to reorganise attention patterns toward proverbial semantics. TF-IDF, by contrast, captures category-specific lexical regularities statistically without requiring new representational structure. The character n-gram variant (0.214) also performed well, confirming that morphological surface patterns carry genuine discriminative signal for some categories.

The results partially support the hypothesis that language-specific or African-language pretraining yields an advantage. AfriBERTa-Large (0.202 accuracy, 0.109 Macro-F1) outperformed SwahBERT (0.155 accuracy, 0.066 Macro-F1) and both DistilBERT configurations. This suggests that breadth of African-language exposure, since AfriBERTa covers 11 African languages, may matter more than depth of Swahili-specific training for this task. SwahBERT's monolingual news-domain pretraining is likely more stylistically distant from proverbial language than AfriBERTa's broader corpus.

DistilBERT-multilingual's strong epoch sensitivity is also notable: the 3-epoch run (Macro-F1 0.010) is effectively random, while the 10-epoch run (Macro-F1 0.072) approaches SwahBERT's performance. The multilingual model requires substantially more fine-tuning to adapt its broad language representations to the narrow Swahili proverb domain.

The divergent effects of MLM domain adaptation, a small gain for AfriBERTa (+0.004) and a small decline for SwahBERT (−0.012), are the most intriguing finding in the transformer experiments. For AfriBERTa, the result is consistent with the expectation that a single pass over the corpus shifts masked token predictions toward proverbial Swahili distributions. For SwahBERT, the decline may reflect overfitting during the MLM adaptation stage: a 12-layer model adapting on only 5,060 short sequences may shift representations away from generalisable patterns toward corpus-specific surface forms, reducing test-set generalisation.

The inconsistency across models suggests that the benefit of MLM adaptation at this data scale is unreliable. Extended adaptation with learning-rate tuning or regularisation may be necessary to produce consistent gains, which was precluded here by computational constraints.

On LLM Prompting for Low-Resource Tasks

The Llama-3 zero-shot result (0.200 accuracy) is unexpectedly competitive with the best transformer fine-tuning results, though the 20-sample evaluation makes direct comparison unreliable. The few-shot result (0.150) underperforming zero-shot is also unexpected and contrasts with the typical pattern where examples improve performance. One explanation is that 10 illustrated categories out of 56 may have introduced label bias: after seeing examples for a subset of categories, the model may have over-predicted those categories, reducing accuracy on the remaining 46. Prompt and label-

parsing design are known to have outsized effects on LLM classification performance (Brown et al., 2020).

A broader limitation is that the 56 category labels are abstract, culture-embedded thematic concepts (Pride and Humility, Speech and Silence, Patience and Impatience) that may not be well represented in Llama-3's pretraining distribution in a form that enables reliable zero-shot alignment. This is a qualitatively harder problem than standard topic classification.

6. Conclusion

This study evaluated four NLP paradigms on the classification of 5,060 Swahili proverbs into 56 semantic categories: static embedding classification, transformer fine-tuning, MLM domain adaptation, and LLM prompting. The task is representative of a broader class of NLP problems characterised by small labelled datasets, severe class imbalance, metaphorical language, and a morphologically rich target language underrepresented in standard pretraining corpora.

The central empirical finding is that TF-IDF combined with a shallow neural network achieved the highest test accuracy (0.240), outperforming all transformer and LLM approaches. The best transformer result, AfriBERTa-Large with MLM domain adaptation, reached 0.206. MLM adaptation produced a small positive gain for AfriBERTa (+0.004) but a slight decline for SwahBERT (−0.012), suggesting the technique is unreliable at this data scale. Llama-3-8B zero-shot achieved 0.200, unexpectedly outperforming few-shot prompting (0.150). All methods substantially exceeded the random baseline (roughly 0.018) but remain far from practical utility, underscoring the inherent difficulty of metaphor-grounded semantic classification from limited labelled data.

This work makes three primary contributions. First, it provides the first systematic comparison of static embeddings, BERT-style fine-tuning, domain-adaptive pretraining, and LLM prompting on Swahili proverb classification, establishing reproducible baseline performance figures for a previously unstudied task. Second, it demonstrates the practical importance of metric selection for imbalanced classification: where Macro-F1 is available it tells a substantially different story from accuracy, a point that generalises to any fine-grained multi-class task with long-tail support. Third, it provides a reproducible experimental framework tracked via Weights & Biases and MLflow that can be extended to other low-resource African language classification tasks.

Limitations

Class imbalance and low-support categories with fewer than 10 training examples constrain what any current method can reliably learn. MLM domain adaptation was run for only 1 epoch due to computational constraints, and extended adaptation may produce larger gains. Task 4's evaluation on 20 samples is insufficient for reliable metric estimation and should be treated as exploratory. Finally, the corpus was collected from a single source (Scheven, 2010), which may not fully represent the diversity of Swahili proverbial expression across regional dialects and registers.

Future Work

Several directions offer promising extensions. Data augmentation, including back-translation through a high-resource language, morphological variation, or paraphrase generation, could expand the effective training set for low-support categories. Parameter-efficient fine-tuning methods such as LoRA (Hu et al., 2022) would make it feasible to adapt larger LLMs within standard GPU memory constraints. Cross-lingual transfer from structurally related Bantu languages such as Zulu, Xhosa, and

Lingala could provide useful inductive bias. Extending the study to proverb datasets in other low-resource languages would test the generalisability of these findings beyond Swahili.

GitHub Repository:

<https://github.com/belinatom/NALAPROJECT>

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Note on Use of AI Tools

Large language model assistance (Claude, Anthropic) was used during the preparation of this work for two purposes: editorial refinement of written content and code development support across pipeline

components. All research design, methodological decisions, empirical analysis, and interpretations remain solely mine.

Appendix

A. Label Distribution

Figure A1 shows the class distribution across all 56 semantic categories, sorted by frequency. The distribution is heavily skewed: the most frequent categories each contain over 150 samples, while the least frequent contain as few as 5. This imbalance motivates Macro-F1 and MCC as supplementary metrics throughout, as they weight all classes equally regardless of support.

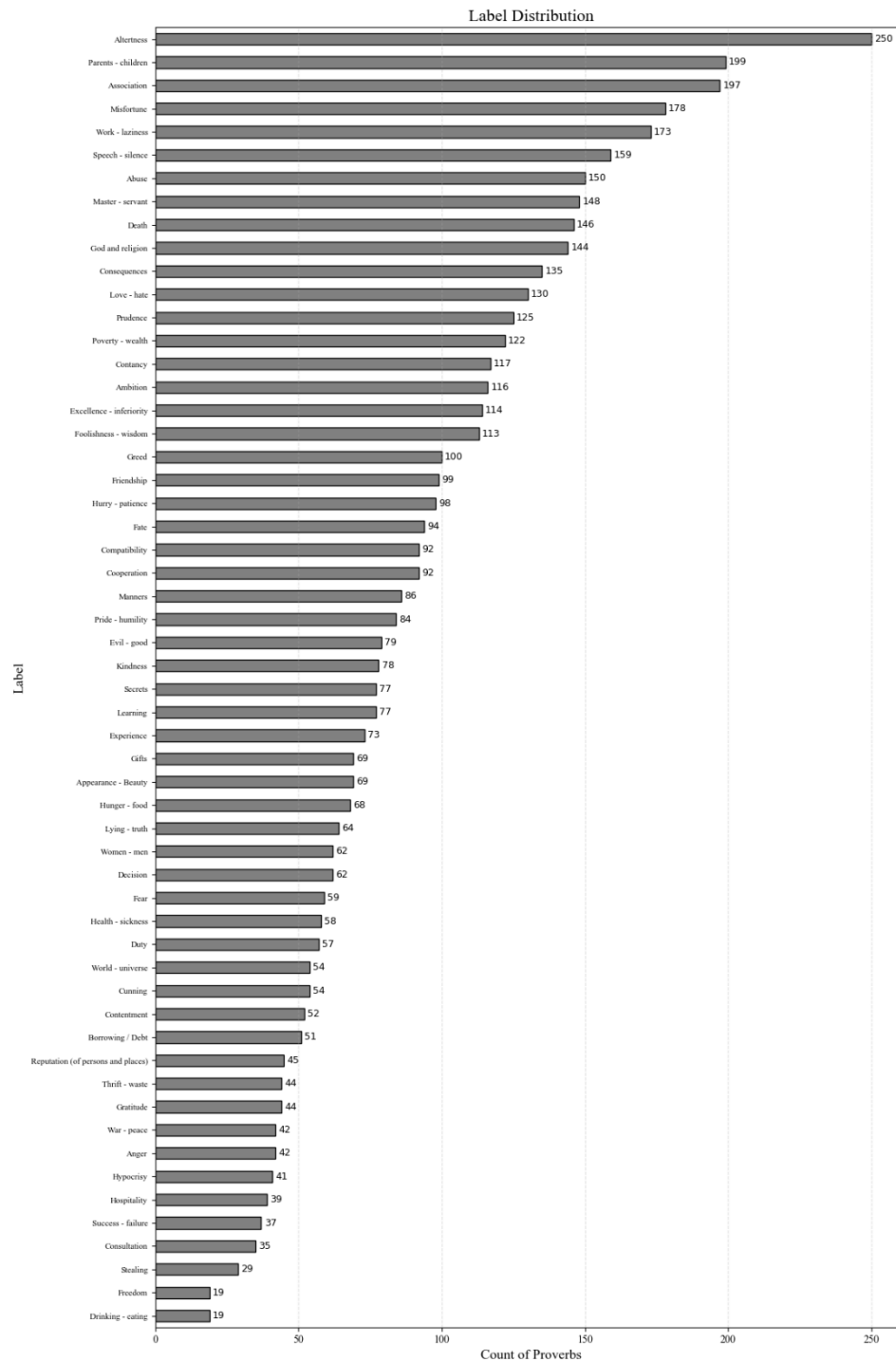


Figure A1. Sample counts per semantic category, sorted by frequency.

B. Per-Class Classification Reports

Full per-class precision, recall, F1, and support metrics for the best-performing method in each task are logged to the Weights & Biases project (nalapro-swahili) and MLflow. Low-support categories (fewer than 10 examples) show near-zero F1 across all methods, consistent with the expectation that no current approach reliably learns from so few examples under 56-class competition.

